2D AI Defect Inspection USER MANUAL

(Version 3.0)

TE CONNECTIVITY

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Outline

2D AI Defect Inspection 3

Before Deployment 4

Deployment Steps..... 5

Select Tool for Your Applications..... 6

Step 1: Image Collection

- Manual Image Collection 7
- InlineImage Collection 10

Step 2: Image Segmentation

- Image Segmentation for Aligned Product 13
- Image Segmentation for Rotated Product..... 14

Step 3: Image tag

- Image labeling for classification detection and anomaly detection 17

Step 4: Training model

- The model is trained on the WEB side..... 18
- Train the model through the client 22

Step 5: Deploy using trained models

- Manual detection deployment 28
- On-line detection deployment.....29

Maintenance, Warranty & Copyright.....32

Contact Us.....34

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2D AI Defect Inspection

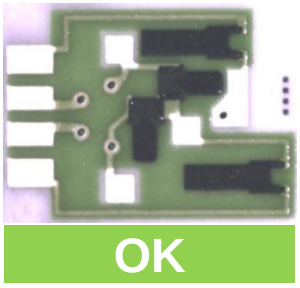
AI Machine Vision (AIMV) System

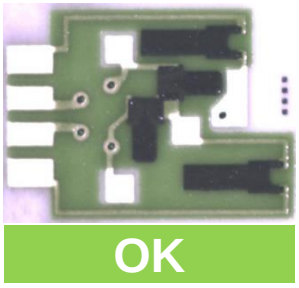
1 Camera System

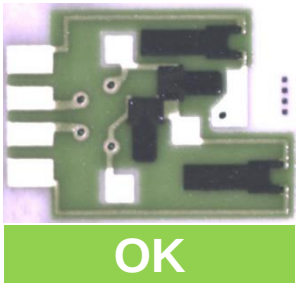
2 AIMV Software on Local PC

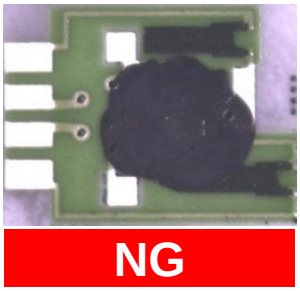
3 AI Model Training on Cloud Service

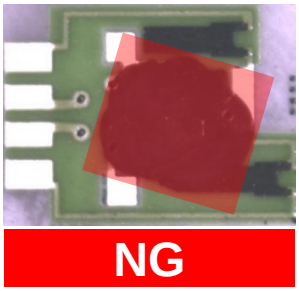
Functionalities

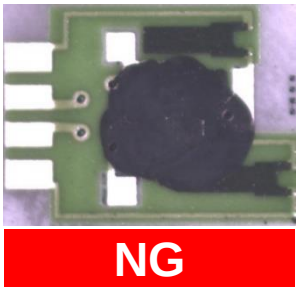












Classification
(Train AI Model based on OK and NG samples)

Defect Location
(Train AI Model based on marked NG samples)

Anomaly Detection
(Train AI Model based on OK samples)

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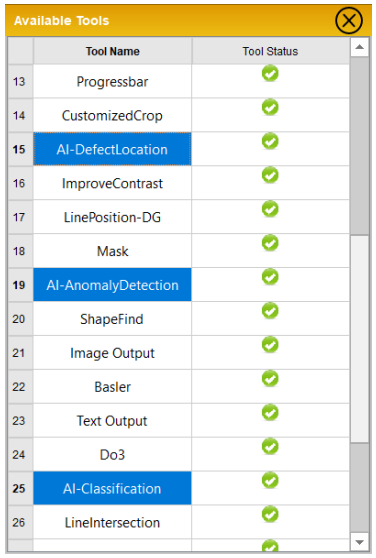
Before Deployment

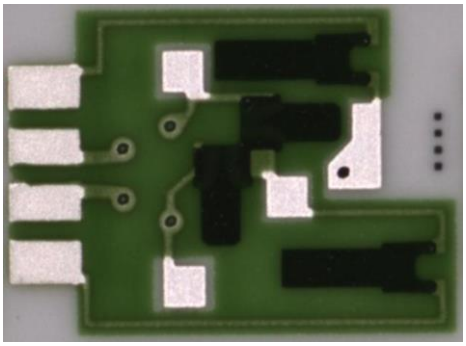
Please read the Quick Start and make sure the items listed below are completed

✓ **1** Microscope Assembled

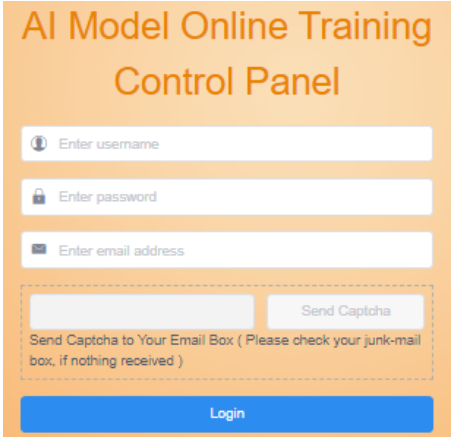
✓ **2** PC with Recommended Configuration



✓ **3** AIMV Software and AI Environment are Installed in the PC



✓ **4** Camera is Registered and Can Capture Clear Image



✓ **5** With Access to AWS Training Server

Recommended Configuration for PC

- CPU: Intel i5 8th series or better
- GPU: NVIDIA GTX 2060 6G or better
- ROM: >= 16 G
- Volume: >= 512 G, SSD is preferred
- USB 3.0
- USB ports > 3
- Windows 10 and above, 64 bit
- Screen >= 21 inches
- Resolution >= 1080P
- Color display

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Deployment Steps

Deployment Steps for Classification and Anomaly Detection

Step 1
Image Collection

↓

Step 2
Image Segmentation

↓

Step 3
Image Labeling

↓

Step 4
Train Model by Training Server

↓

Step 5
Deployment with Trained Model

Need higher accuracy?
Please refer to FAQ

Deployment Steps for Defect Location

Step 1
Image Collection

↓

Step 2
Image Segmentation

↓

Step 3
Image Marking

↓

Step 4
Train Model by Training Server

↓

Step 5
Deployment with Trained Model

Need higher accuracy?
Please refer to FAQ

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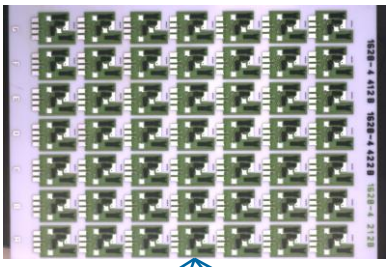
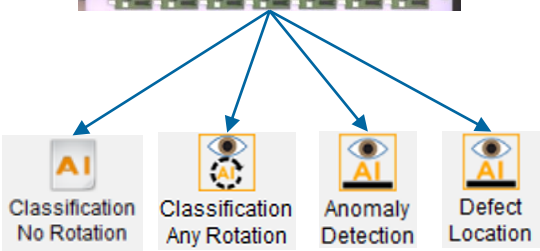
Select Tool for Your Applications

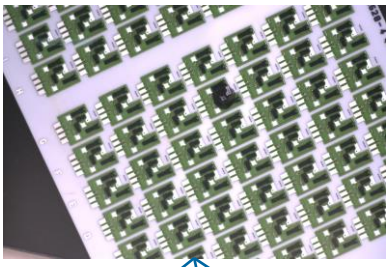
Selectable Tools for Different Product Types

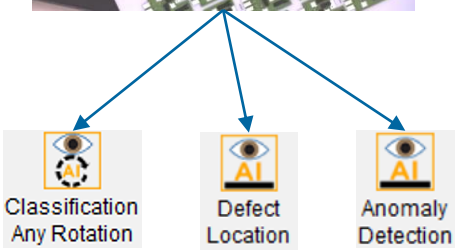
Horizontally Aligned Product
(With Faster Processing Speed)



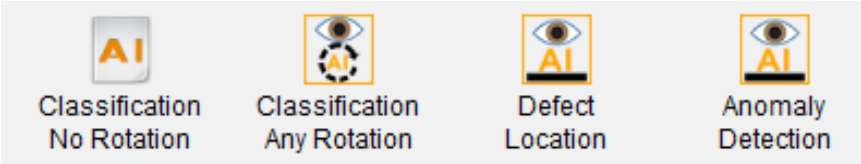


Rotated Product





Create Your Application



1. Click required tool

2. Select “Directory” and input “Folder Name”
3. Click “Ok”, your application is created and the software will start Image Collection project. Please refer to Image Collection page for more operations on image collection

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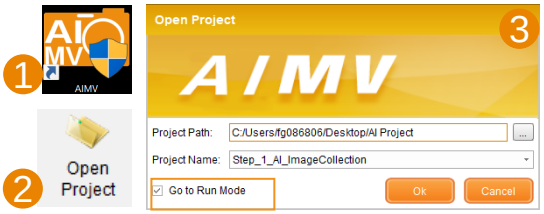
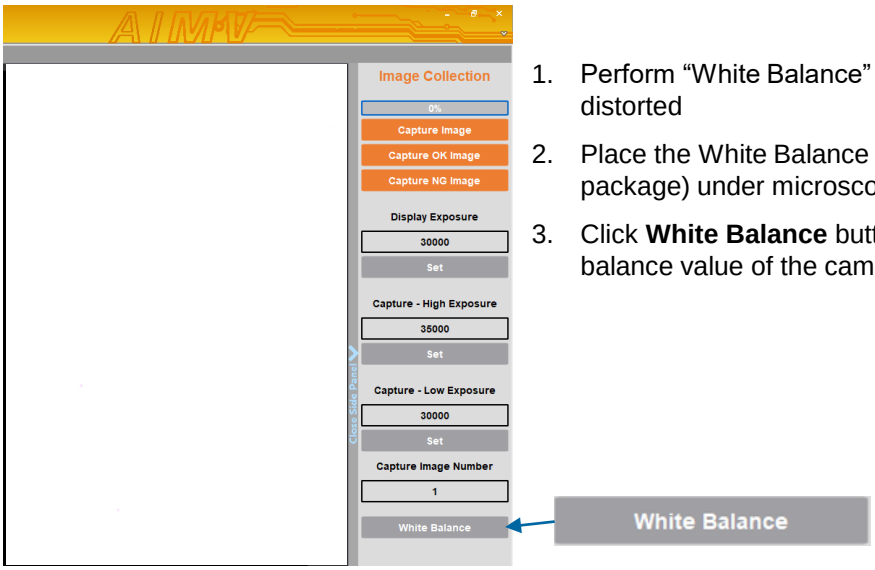
Manual Image Collection

Open Image Collection Project



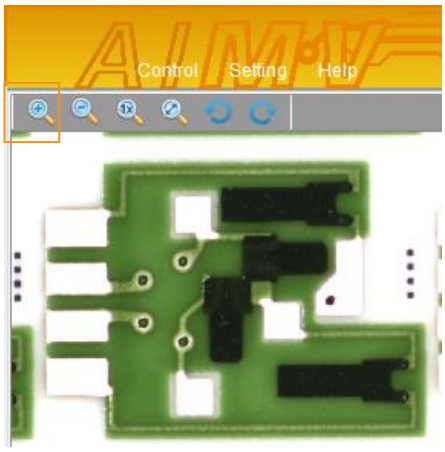
1. Open AIMV Software
2. Click “Open Project”
3. Open Project “Step_1_AI_ImageCollection”, and make sure “Go to Run Mode” is checked

White Balance if Necessary



1. Perform “White Balance” if the color of image is distorted
2. Place the White Balance Board (accessory in the package) under microscope
3. Click **White Balance** button to adjust the white balance value of the camera

Capture Clear Image of Product



1. Click the zoom button to have a zoomed view of product

2. Rotate to adjust the height of the camera

3. Rotate to adjust the lens focus

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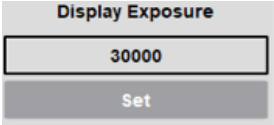
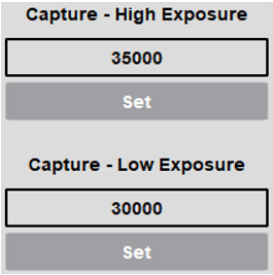
Manual Image Collection

Set Exposure Value and Capture Image Number



- Set “Display Exposure” for real-time image display when image collection
- Click “Set”, then the effect of current exposure will be displayed

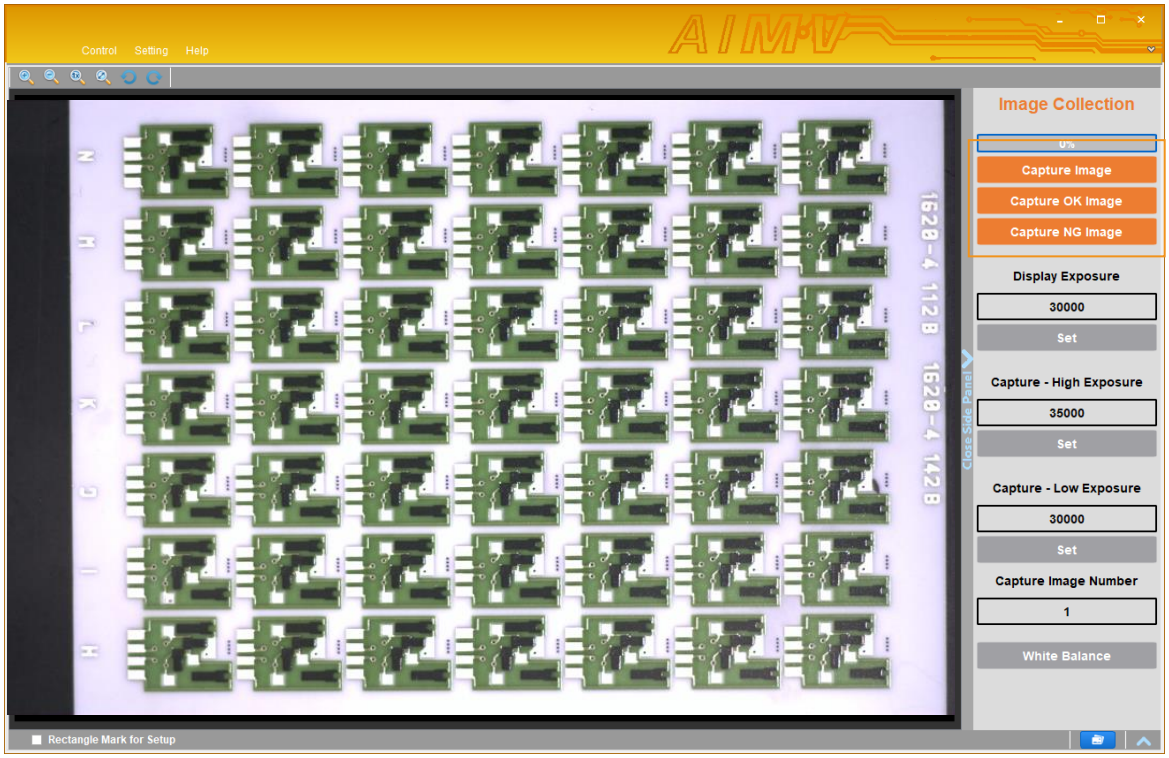


- Set “Capture – High Exposure” and “Capture – Low Exposure”, the camera exposure of the captured image will be between those two values
- Click “Set”, then the effect of current exposure will be displayed

- Set “Capture Image Number” to define how many images will be captured and saved for one capture

- Refer to FAQ for any question

Capture Product Image



1. Place product under the camera
2. Click “Capture Image”, “Capture OK Image” and “Capture NG Image” to capture the images

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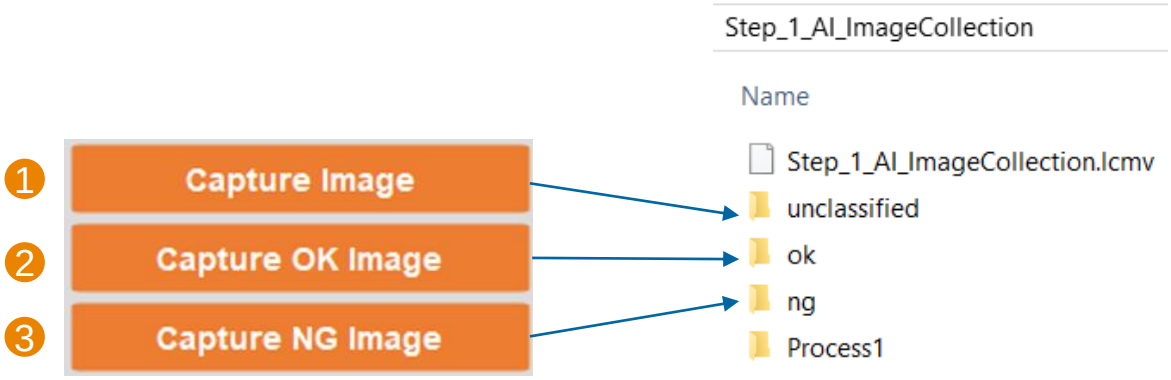
Manual Image Collection

Image Saving Folder



1. Click “Capture Image”, the image will be saved in the “unclassified” folder of current project folder
2. Click “Capture OK Image”, the image will be saved in the “ok” folder of current project folder
3. Click “Capture NG Image”, the image will be saved in the “ng” folder of current project folder

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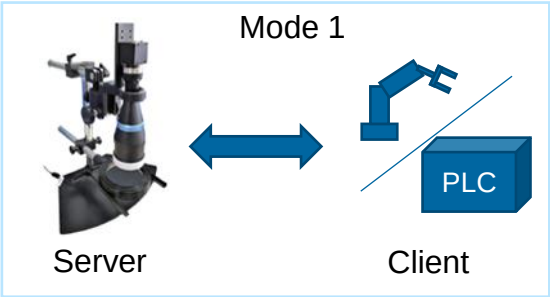
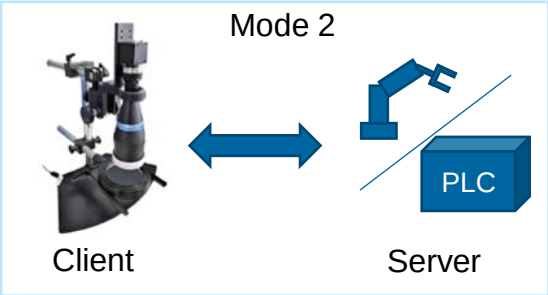
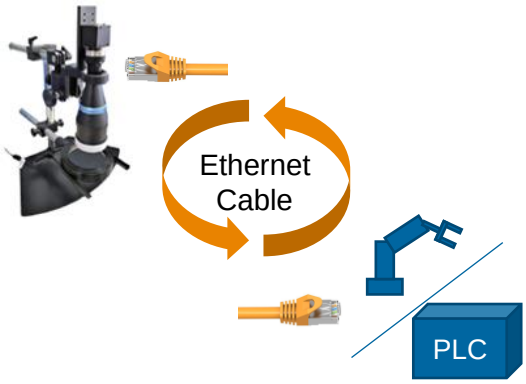
TCP Communication with PLC/Robot

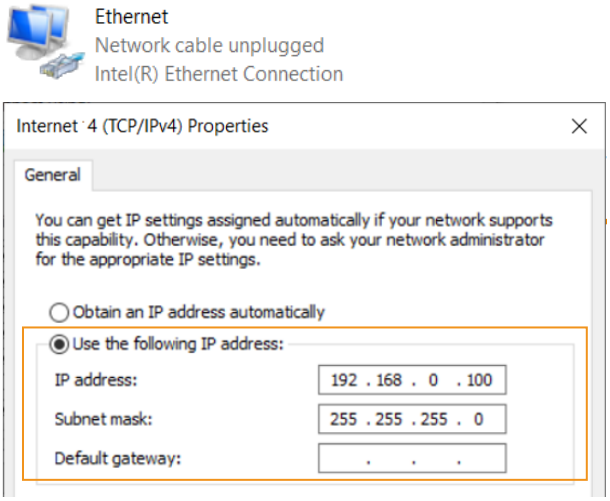
- AIMV software can communicate with PLC/Robot through TCP communication
- Function of TCP Communication
  - Receive trigger from PLC/Robot
  - Send inspection result to PLC/Robot
- Support two kinds of work mode
  - Mode 1: AIMV software works as TCP Server, PLC/Robot works as TCP Client
  - Mode 2: AIMV software works as TCP Client, PLC/Robot works as TCP Server





Hardware Preparation





1. AIMV PC is connected with PLC/Robot by Ethernet Cable
2. The IP address settings of AIMV software and PLC/Robot are in the same network
3. The PLC/Robot supports TCP communication
4. The proven brands are Allen-Bradley(AB), Siemens, Omron, ABB, EPSON, UNIVERSAL ROBOTS (UR), etc.

Software Preparation

1. Please go through the **Manual Image Collection** to make sure the camera is able to capture product image correctly

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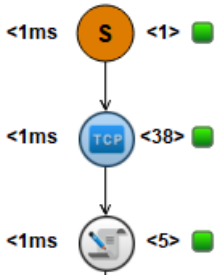
Inline Image Collection

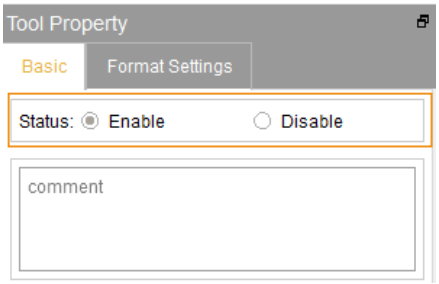
Open Image Collection Project

1. Open AIMV Software
2. Click “Open Project”
3. Select project “Step_1_AI_ImageCollection”
4. If the following step is preformed, make sure “Go to Run Mode” is checked to start inline image collection, or else, make sure Go to Run Mode” is unchecked to open the editor mode

Enable PLC/Robot Trigger by TCP

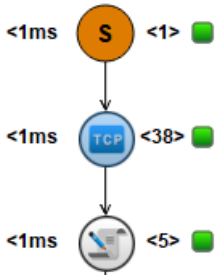


1. Click TCP after S

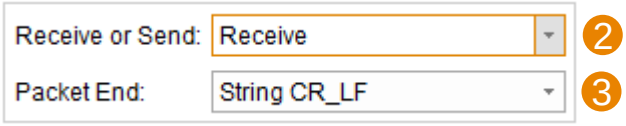


2. Change the tool Status to **Enable** to enable the TCP communication

Align the TCP Settings with PLC/Robot

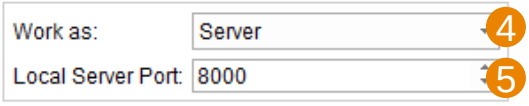


1. Click TCP after S



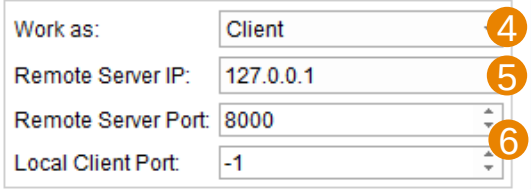
2. Set **Receive or Send** to **Receive** to receive trigger from PLC/Robot
3. Make sure the **Packet End** is the same as that of PLC/robot

AIMV Software works as TCP Server



4. Set the **Work as** to **Server**
5. Set the **Local Server Port** to number between 1001 to 65536, for example 8000
6. Align the TCP settings of PLC/robot with the settings above

AIMV Software works as TCP Client



4. Set the **Work as** to **Client**
5. Set the **Remote Server IP** to PLC/robot IP address
6. Set **Remote Server Port** and **Local Client Port** per the settings of PLC/robot
7. Align the TCP settings of PLC/robot with the settings above

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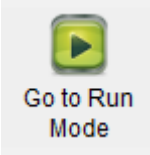
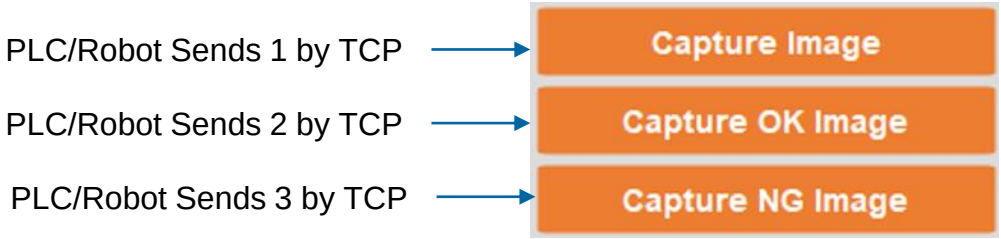
Inline Image Collection

Start Inline Image Collection



1. Click “Go to Run Mode” and save the settings to start the Run Mode

PLC/Robot Triggers to Collect Image

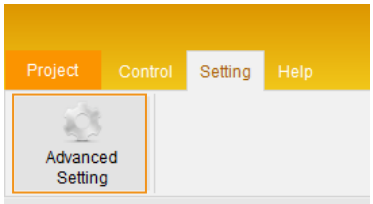


1. If the PLC/Robot Sends 1 by TCP, it’s the same function as click “Capture Image” button
2. If the PLC/Robot Sends 2 by TCP, it’s the same function as click “Capture OK Image” button
3. If the PLC/Robot Sends 3 by TCP, it’s the same function as click “Capture NG Image” button

*Please refer to FAQ for connection problem

Auto-Run Inline Image Collection when Opening Software

The function is for the customer want to start Image Collection automatically after opening AIMV software



1. Open AIMV Software
2. Click Advanced Setting in Setting Tab

3. Check “Auto Open Project when Startup”
4. Select the Project Folder
5. Select the Project Name to “Step_1_AI_ImageCollection”, click OK

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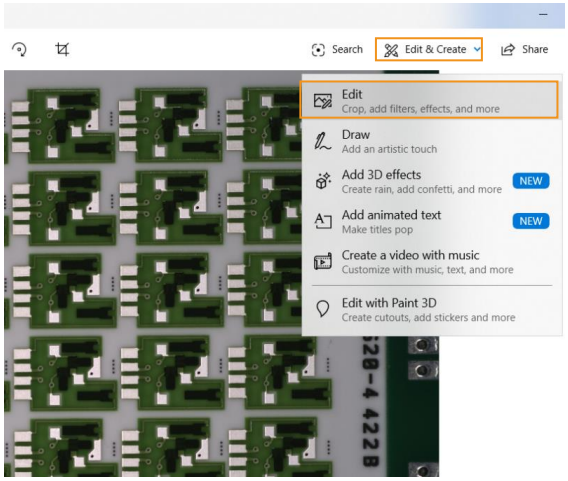
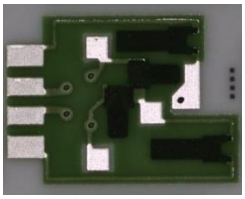
Image Segmentation for Aligned Product

Generate Image Template





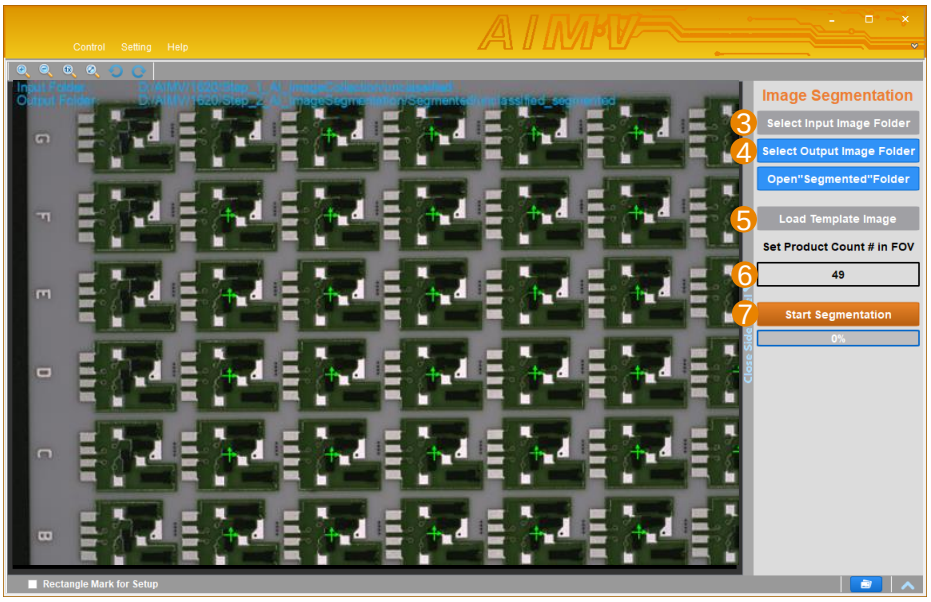
Template.jpg

1. Use Win10 Photo to Segment One Product Image as the Template. If you are not familiar with Win 10 Photo program, please refer to FAQ
2. Make sure the image is saved in jpg format

Open Image Segmentation Project

1. Open AIMV Software
2. Click “Open Project”
3. Open “Step_2_AIC_ImageSegmentationNoRotation” project, and make sure “Go to Run Mode” is checked

Start Segmentation



3. Click “Select Input Image Folder” to select folder containing the captured images
4. Click “Select Output Image Folder” to select folder to save segmented images
5. Click “Load Template Image” to load template image segmented by Win 10 Photo program
6. Set the “Procut Count # in FOV”, which means how many products within one image
7. Click “Start Segmentation” to start, the segmented images will be saved in the output folder

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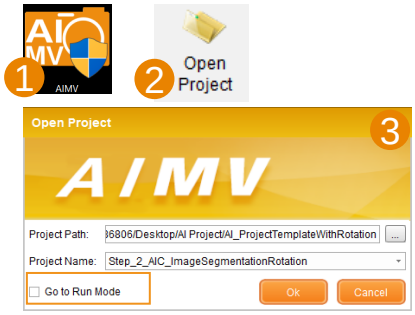
Image Segmentation for Rotated Product

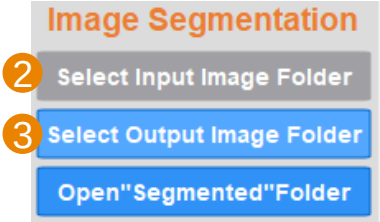
Open Image Segmentation Project



1. Open AIMV Software
2. Click “Open Project”
3. Open Project “Step_2_AIX_ImageSegmentationRotation”, and make sure “Go to Run Mode” is unchecked
   - Step_2_AIC_ImageSegmentationRotation for “Classification Any Rotation”
   - Step_2_AID_ImageSegmentationRotation for “Defect Location”
   - Step_2_AIA_ImageSegmentationRotation for “Anomaly Detection”

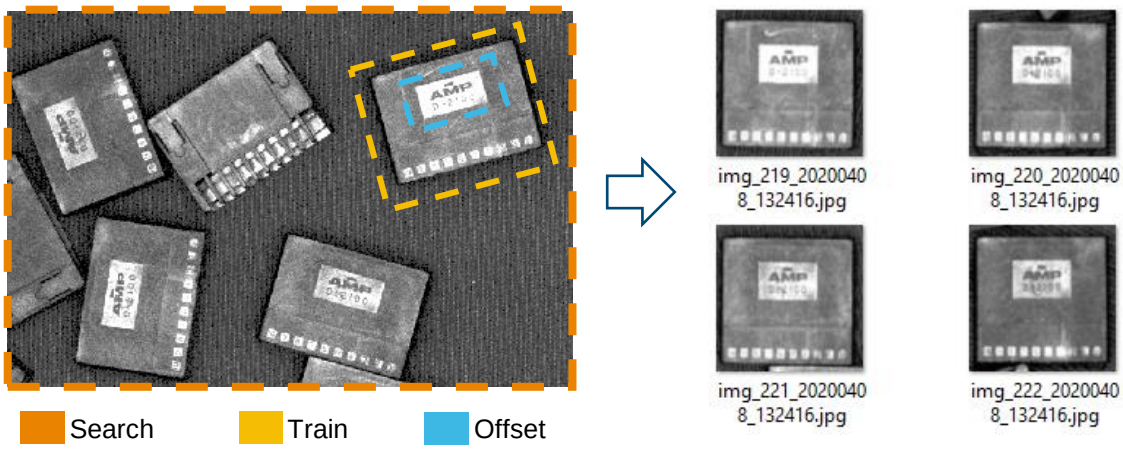
Select Input Image Folder and Output Image Folder

1. Click “Open Side Panel” to open the side control panel



2. Click “Select Input Image Folder” to select folder to save segmented images
3. Click “Select Output Image Folder” to select output image folder for segmented images

Train Rectangle, Search Rectangle and Offset Rectangle



- Search rectangle defines the whole area to search the product
- Train rectangle defines the feature to locate the product
- Offset rectangle defines the output area of the product

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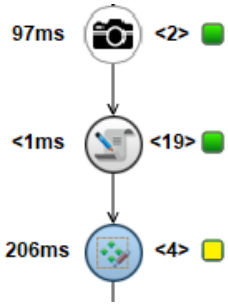
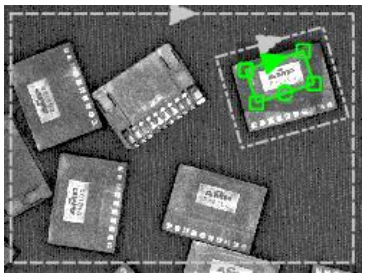
Image Segmentation for Rotated Product

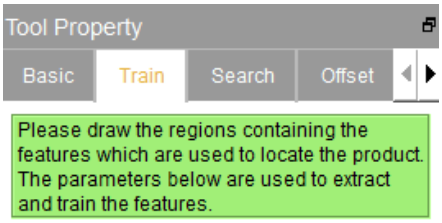
Define Train Rectangle



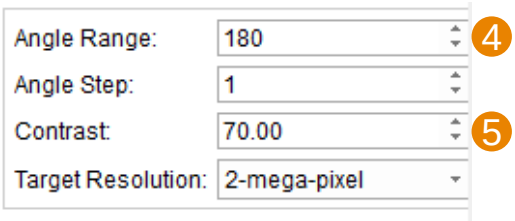
1. Click Node



3. Drag the rectangle to cover the feature to locate the product

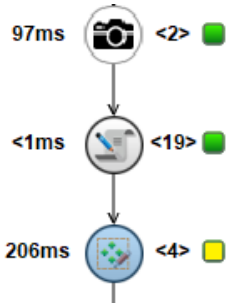


2. Switch to “Train” tab

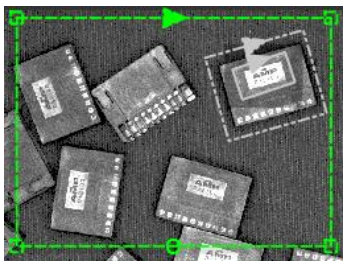


4. Set “Angle Range” to specify the rotation angle of the product. 180 means range is between -180 degrees to 180 degree. Reducing the range can speed up the location
5. Set “Contrast” to define the edge contrast within the rectangle
6. Click Train to train one location template

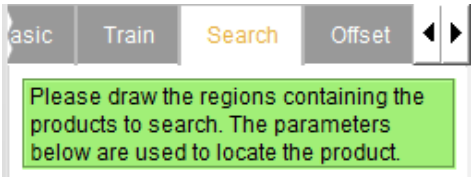
Define Search Rectangle



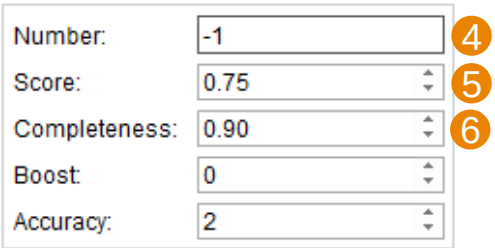
1. Click Node



3. Drag the rectangle to cover the area to search the product



2. Switch to “Search” tab



4. Set “Number” to specify product number to search, -1 means find all possible products
5. Set “Score” to control the similarity between found product and trained template. 0.75 means at least 75% similarity is required for one found product
6. Set “Completeness” to control the completeness of one found product. 0.90 means at least 90% completeness

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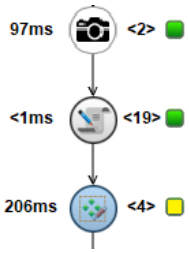
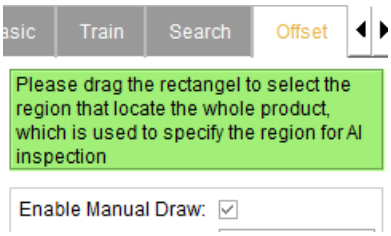
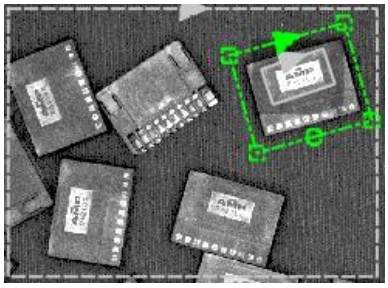
Image Segmentation for Rotated Product

Define Offset Rectangle

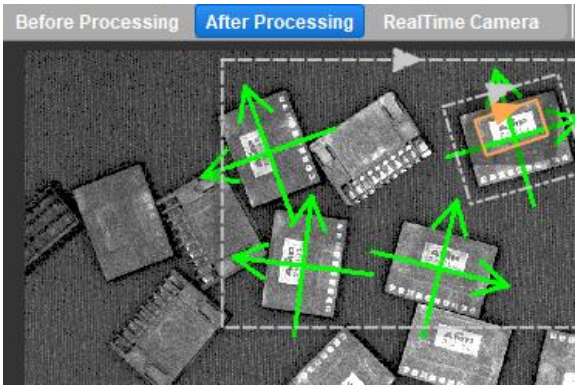




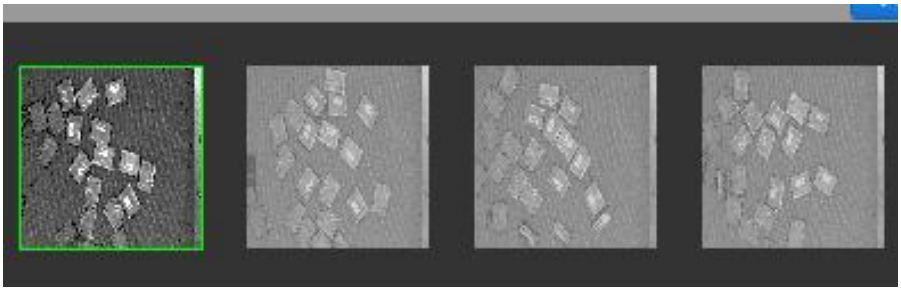


1. Click Node
2. Switch to “Offset” tab, check the “Enable Manual Draw”
3. Drag the rectangle to cover the output area of the product

Test Location Effect

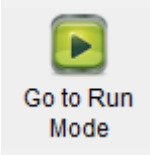


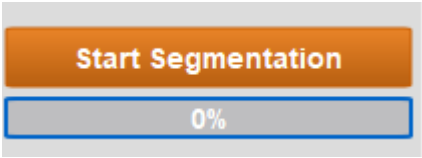
1. Click “After Processing”, the location result will be marked by green arrow

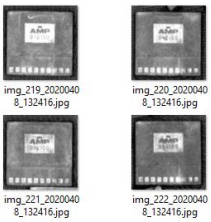


2. Switch between images to see the effect, if not good, adjust settings in previous steps. Refer to FAQ for any question

Start Segmentation







1. Click “Go to Run Mode”
2. Click “Start Segmentation”
3. The segmented images will be saved in the output folder

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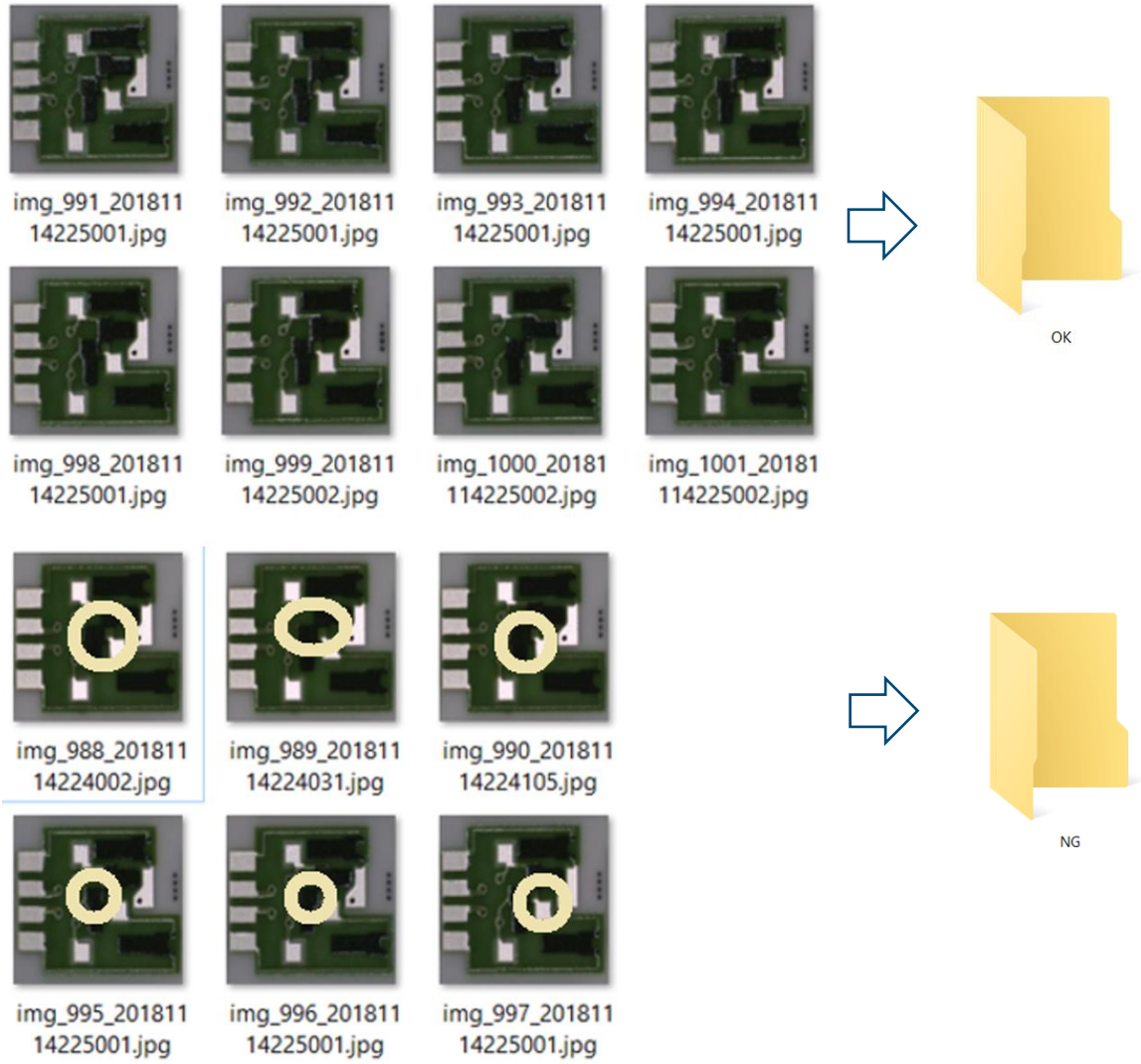

1. Create two folders named “OK” and “NG”
2. Manually classify the segmented images into OK or NG folder per the product quality

Rules for Labelling

- Make sure each picture in “OK” folder is one OK product, and each picture in “NG” folder is one NG product
- Make sure the ratio of NG samples > 20% of total samples. For example, if there are 200 pictures in OK folder, 100 pictures in NG folder, then the ratio of NG samples is 100/(200+100) = 33.3%, which is > 20%
- If the ratio of NG samples is < 20%, collect more NG samples

- Tip: This operation can be performed only on the web client.

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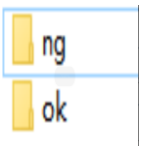
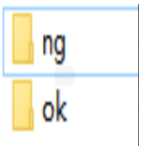
The model is trained on the WEB side

Organize the Training Data

1. Create one folder named *1620* (Name is up to you)
2. Within the folder, create two folders, testing_data and training_data
3. Copy Data to a folder

Classification		In two subfolders ok: store the results for ok picture ng: store the results for the ng pictures
Defect Location	Create a single folder	A tagged image exists in the folder
Anomaly Detection		In two subfolders ok: store the results for ok picture ng: store the results for the ng pictures

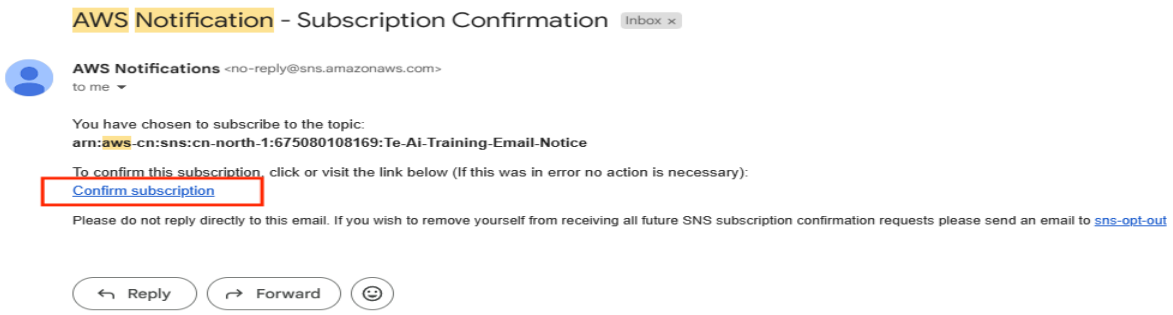
4. Zip the folder

1620.zip

Login to The AWS Training Server

1. Go to website:
   - Users outside China: xxx
   - The preferred browsers are Chrome
   - Enter the user name, password and e-mail address, and click "Send Captcha"
2. Find the captcha in your mailbox, enter the captcha, and click login

TIP: If you login for the first time, you need to find the corresponding email in the email box and click subscribe. Only after successfully subscribing can you login to the service normally, as shown below:



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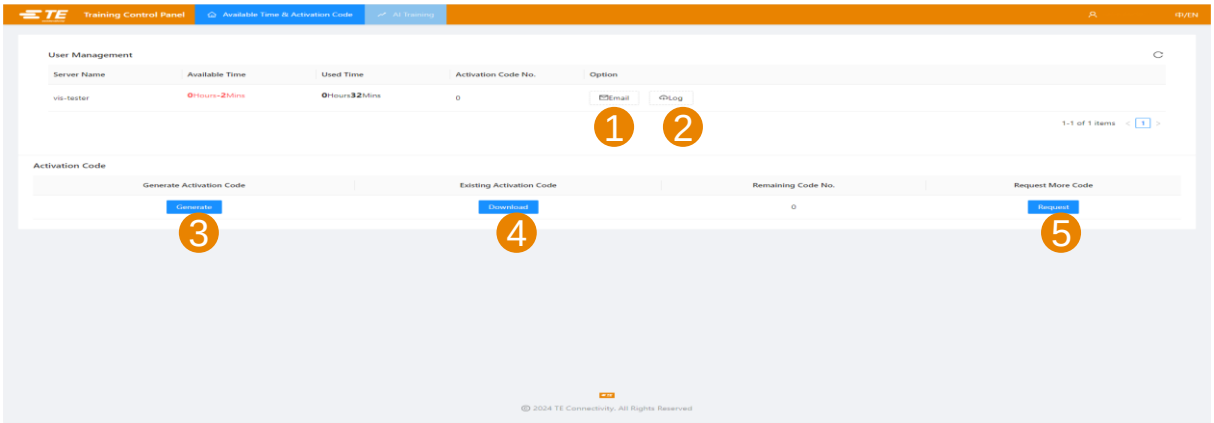
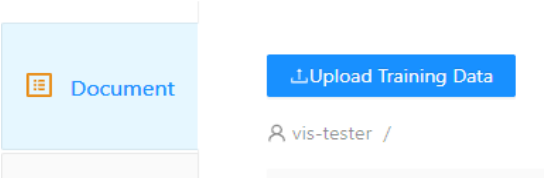
The model is trained on the WEB side

Training duration and activation code management



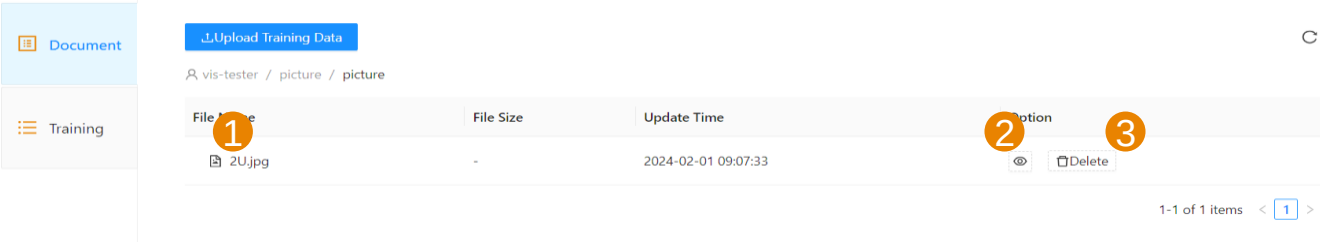
1. Click "Mailbox information", the mailbox information page pops up, you can bind and unbind the mailbox
2. Click "Log" to download the log information of last month or this month
3. Click "Generate" to enter the generate activation code page, enter the camera code and factory, and click OK
4. Click "Download" to download the generated activation code information
5. If you want to apply for more activation codes, you can click "Request", the application procedure will be displayed

Upload data to the training server



1. Enter the AI training page and upload the data
2. Click on "Document"
3. Click "Upload Training Data", select the zip file, and then upload the zip file

Download, preview, and delete images



1. Go to the picture page and click the name of the picture to download the picture
2. Click on the small eye （ ） to preview the current picture
3. On the picture page, click to delete （ Delete ） the picture (on the picture folder page, click Delete to delete the entire folder)

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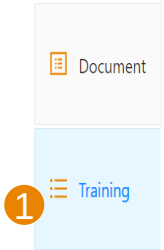
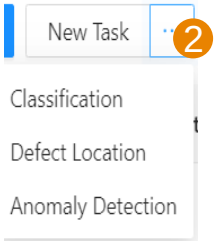
The model is trained on the WEB side

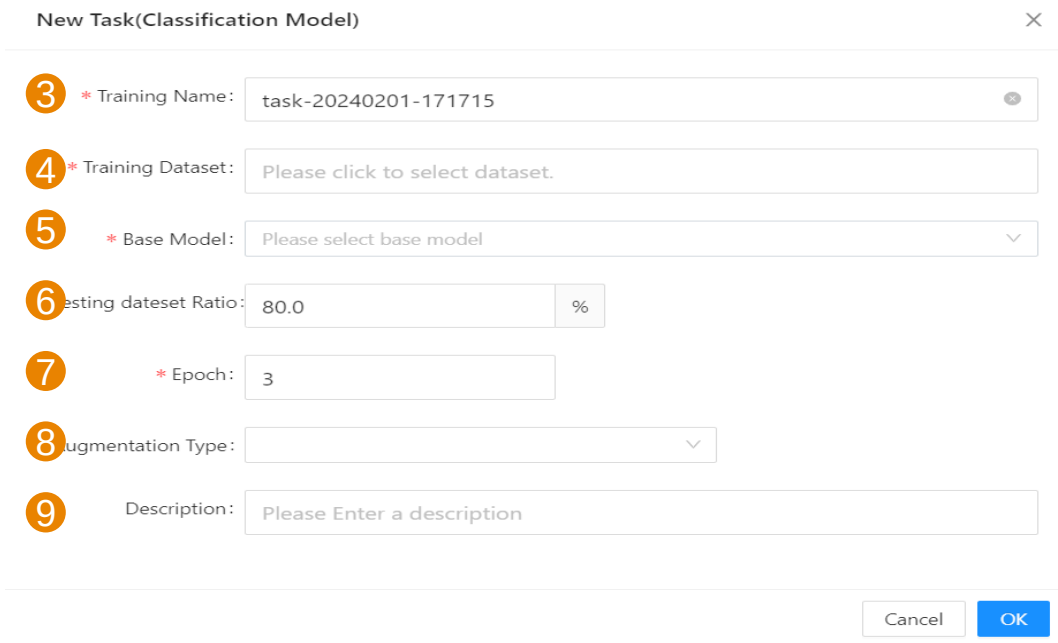
Create a training task





1. Select the "Training" field

2. Click "New Task" and select the training task based on your application



3. Enter a Training Name
4. Click "Training Dataeset" and select the corresponding folder from the uploaded data
5. Select “Base Model” :
   - For a new training task, select 0
   - For continuous training, select the number of the previous model
6. Set the Testing dateset Ratio
7. Enter the Epoch. The default is 3 rounds and the maximum is 1000 rounds
8. Select alignment
9. Input Remarks
10. Click OK to start Training

Tip: Please close the page during training to avoid extra charges after training!

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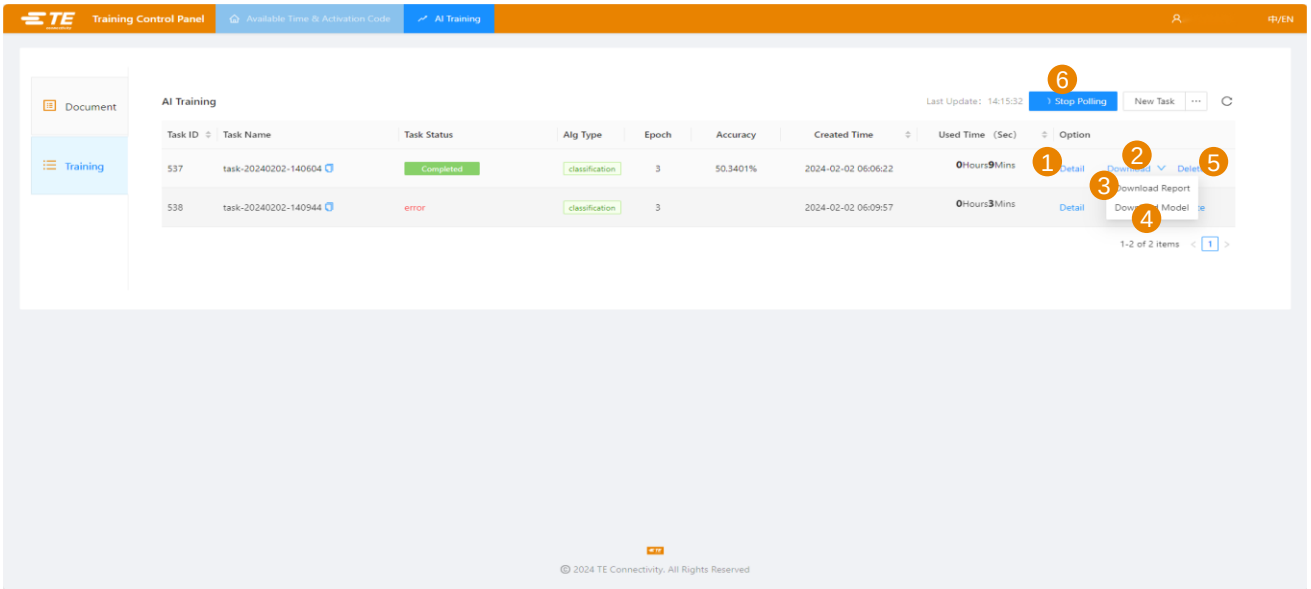
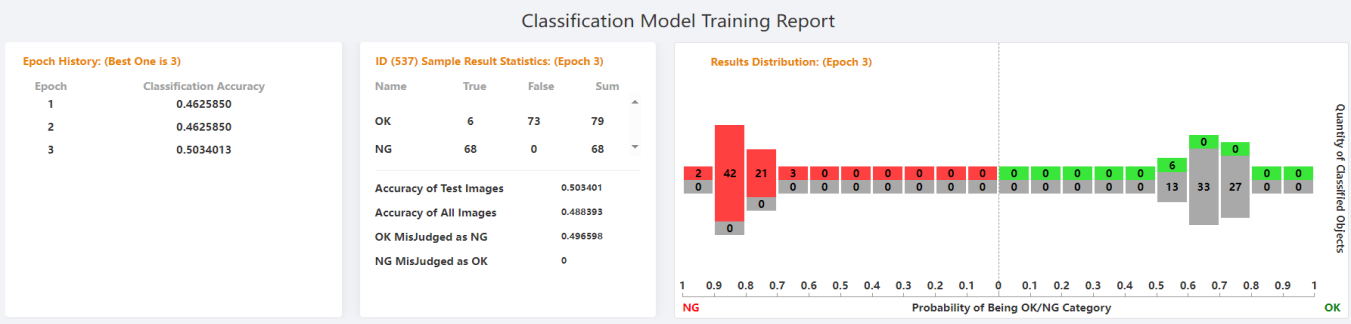
The model is trained on the WEB side

View training parameters, training results reports, and download training models



1. Click on the training parameters to view the training parameters details
2. The training task has been completed. Click download to download the training result report



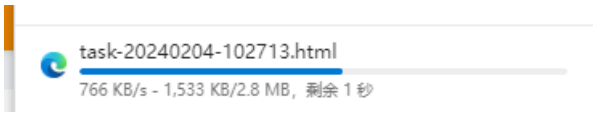
The accuracy of each round during training

Test data in general

* For explanations of training reports and tips on improving accuracy, see the FAQ

3. Click download to download the model, and the page displays the download progress

4. Click Download to download the report, and the page displays the download progress



5. If the task is not in training state, you can delete the task Delete . Do not perform this operation if the task is in training.

6.Click to enable/stop polling. The training task can be updated in real time at the beginning, and the data will stop updating after closing

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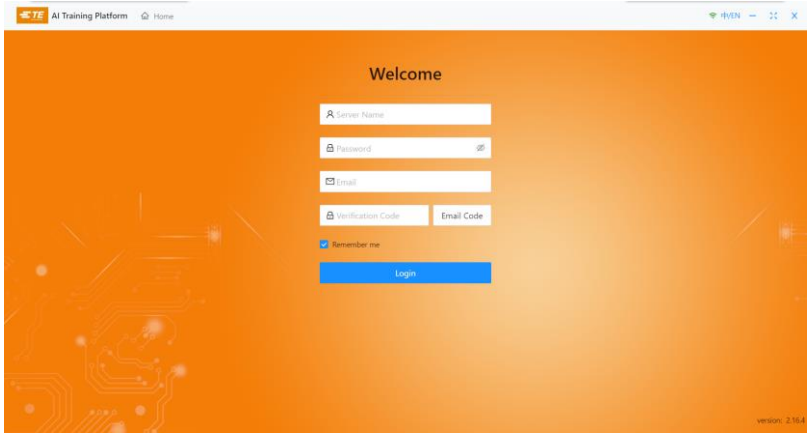
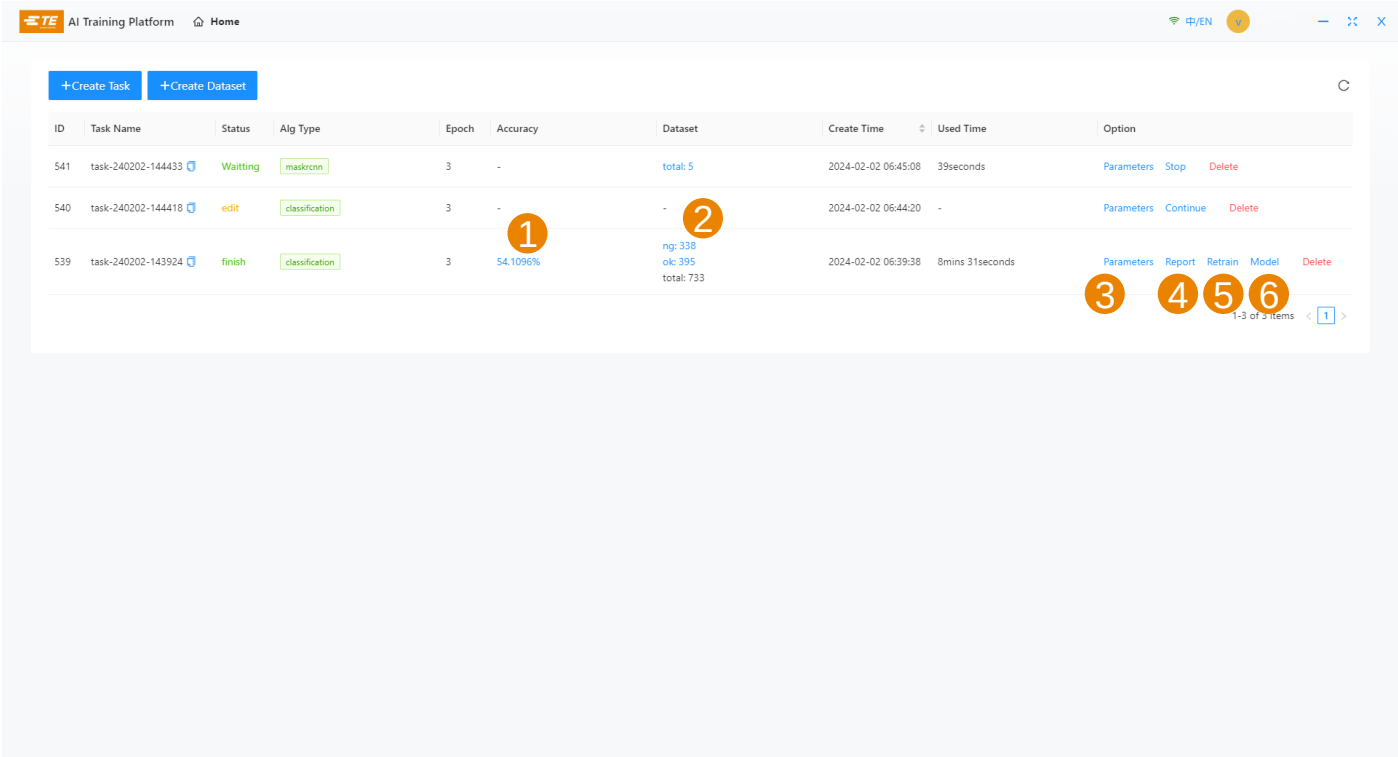
Train the model through the client

Log in to the client training system



1. Double-click to open the installed client
2. Enter your username, password, and email address, then click "Email Code"
3. View the verification code in the mailbox and enter the verification code, then click "Login"
4. If the input information is correct, you can enter the training task list page

Client training task list



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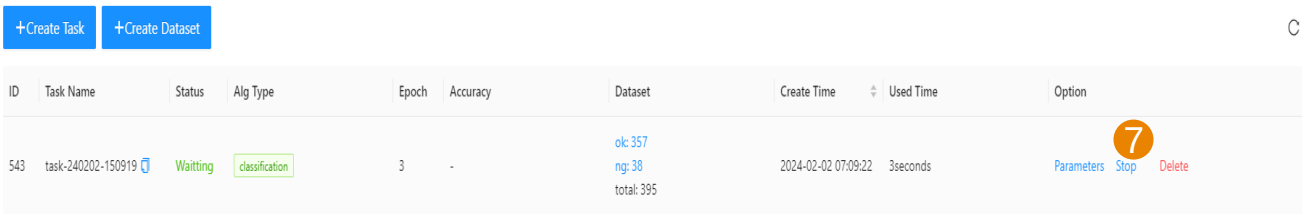
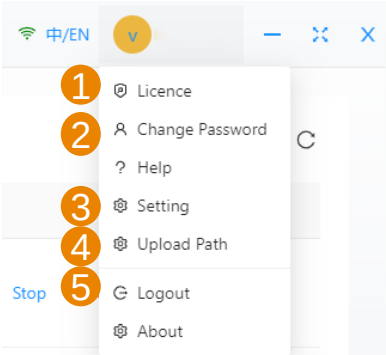
Train the model through the client

1. Click the result data in the precision column to jump to the corresponding training result report page, same as point 4
2. Click the data in the number of pictures column to pop up the group picture data of the corresponding task, and you can select the picture size in the popup window
3. Click "Parameters" to view the details of the training parameters of the corresponding task
4. Click "Report" to download and open the report by default, same as point 1
5. Click "Retrain" to retrain the model. The steps are shown on pages 24-27
6. Click "Model" to download the task model and display the progress. If the download is completed, the interface displays the model path. Click to open the display model storage path
7. Task in training, click "Stop" to stop training



System Settings and training duration, activation code management

Click on the account to pop up a drop-down menu



1. Click "Licence" to manage the training duration and activation code
2. Click "Change Password" to change the user password
3. Click "Setting" to set the training report and the storage path of the model
4. Click "Upload Path" to change the automatic upload path of local pictures
5. Click Logout to return to the login page

Tips：

1. **The automatic upload function only uploads the image data within the last 30 days (image date) in the folder of the set path**
2. **Automatically uploaded image data is saved in the "auto-upload" folder of the Training server**

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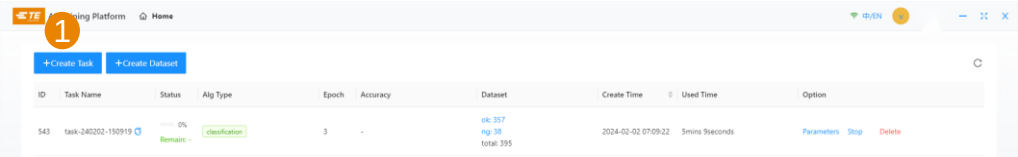
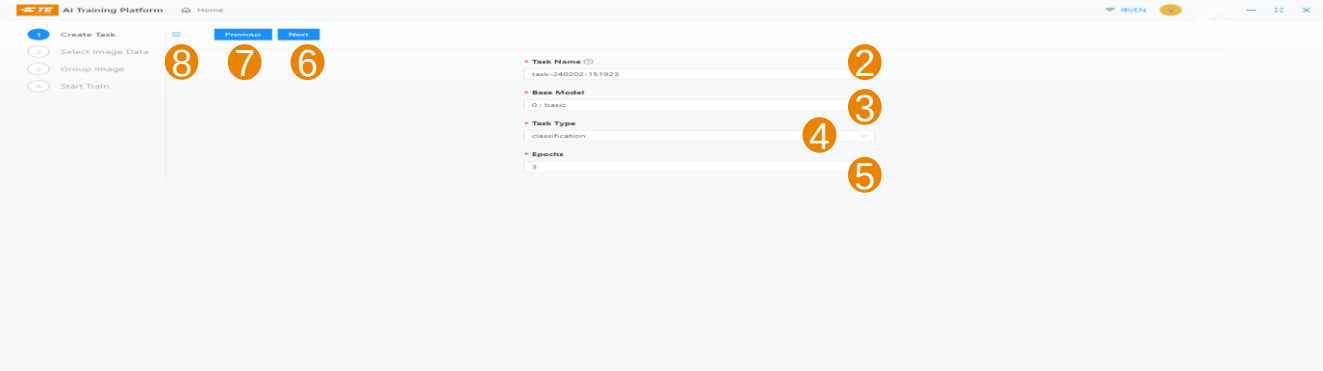
Train the model through the client

Create a training task



1. Click the Create task +Create Task list page to enter the Create training task page



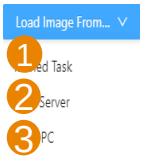
2. The new Task name is automatically generated and can be modified
3. Select "Base Model" :

- For a new training task, select 0
- For continuous training, select the number of the previous model

4. Select the corresponding training task based on your application. The default Task Type is classfication
5. Set training rounds, ranging from 3 to 300 rounds
6. Click to Next enter the Create training data set page
7. Click Previous back to Home page
8. Click to control the sidebar expansion/contraction

Make training data sets

Training data can be loaded from three places



1. Click "Trained task" to enter the page of selecting data set, and click select to load the corresponding data set
2. Click "Training Server" to enter the data set selection page, set the selected data set into groups, and then load the corresponding data set
3. Click "Local PC" to enter the data set selection page, set the selected data set into groups, and then load the corresponding data set
4. You can proceed to the next step only after all data is loaded and the status is successfully uploaded
5. Click the previous step to return to the Create Training task page
6. Clear clear all data that has been uploaded and is in progress

Tip: The training data set of the maskrcnn task does not need to be grouped. Data loaded from the Training server is labeled by default, and data loaded from the local computer is unlabeled by default

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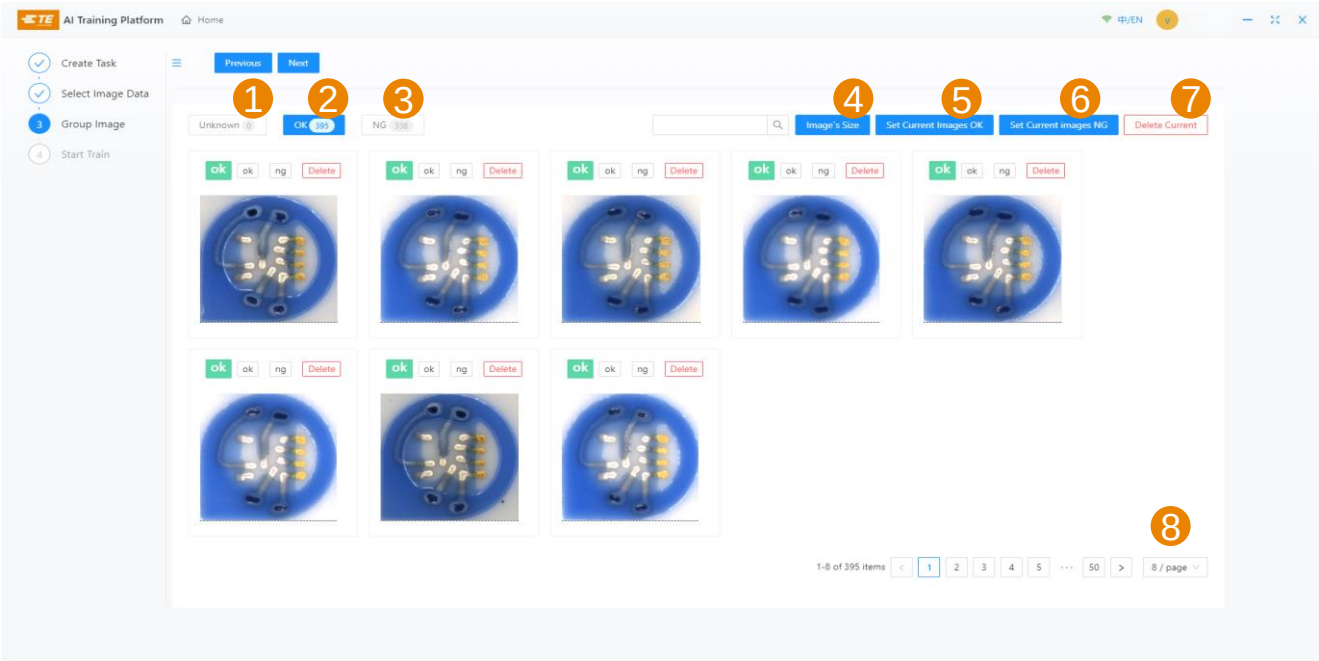
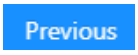
Train the model through the client

Image classification with Oneclass and Classification



1. Click "UnKonwn" to display the uncalibrated data, which can be calibrated
2. Click "OK" to display OK data
3. Click "NG" to display NG data
4. Page picture size display can be modified
5. Batch OK operation can be performed. Batch OK applies to all data on the current page
6. Batch NG operation can be performed. Batch NG applies to all data on the current page
7. You can delete all data on the current page in batches
8. Click to modify the number of pictures displayed on the page
9. After calibration, click Next to enter the training interface

10. Click the previous step, go back to Make the dataset, repeat the previous step to the home page 

11. Move the mouse to the picture, and the picture name is displayed above it
12. The OK, NG, delete operation on the picture is only for the current picture

TIP: Select a single image to calibrate and delete it. The Classification task can proceed to the next step only when OK and NG have data

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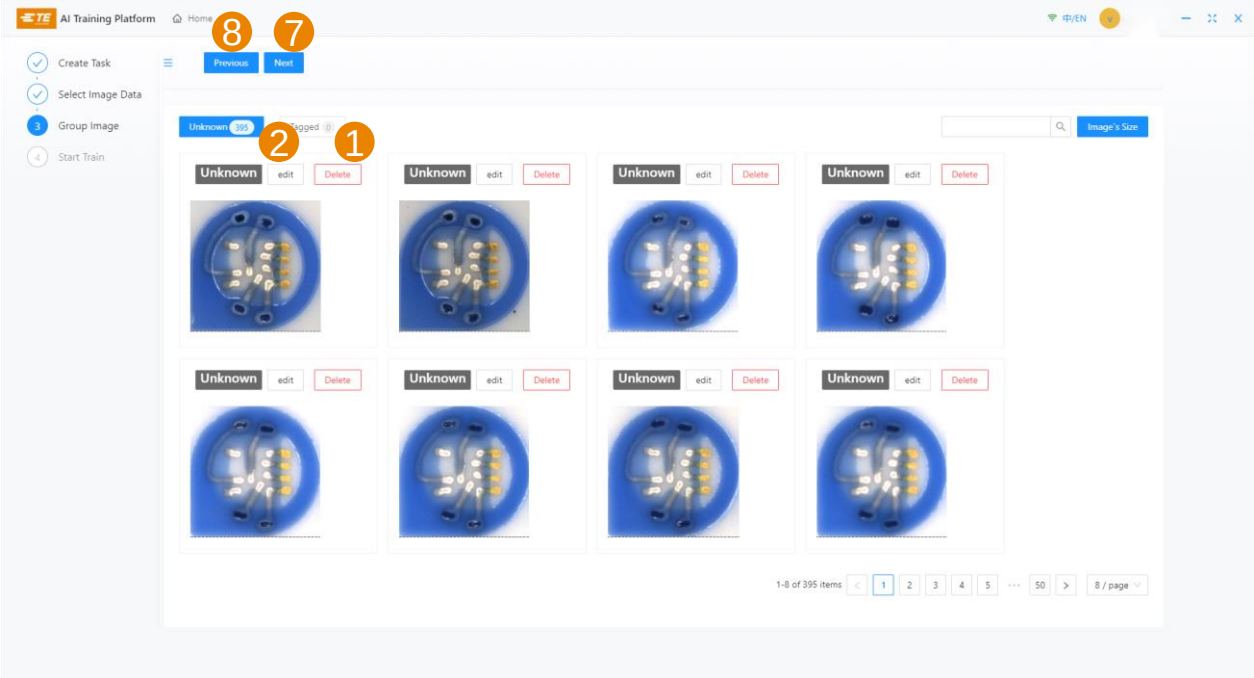
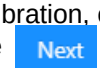
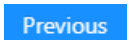
Train the model through the client

Image Classification with Maskrcnn



1. Click "Delete" to delete the selected image
2. Click "edit" to enter the data calibration page. After entering the calibration page, click the left mouse button continuously to calibrate the defect range, and click the right button to automatically seal and end the current defect range calibration

3. Click "Save" to save the calibration information of the image
4. Click "Delete", the calibration box is launched, select the calibration point to be deleted, and confirm that all the selected calibration content in the image is cleared after deletion
5. Click "Revoke" when annotating multiple contents to cancel the previous annotated information point
6. Picture size can be adjusted
7. After all data calibration, click Next to enter the training interface 

8. Click "Previous" and the page will jump back to the "Make Training Data Set" page 

Tip: Proceed to the next step only when calibration data is available

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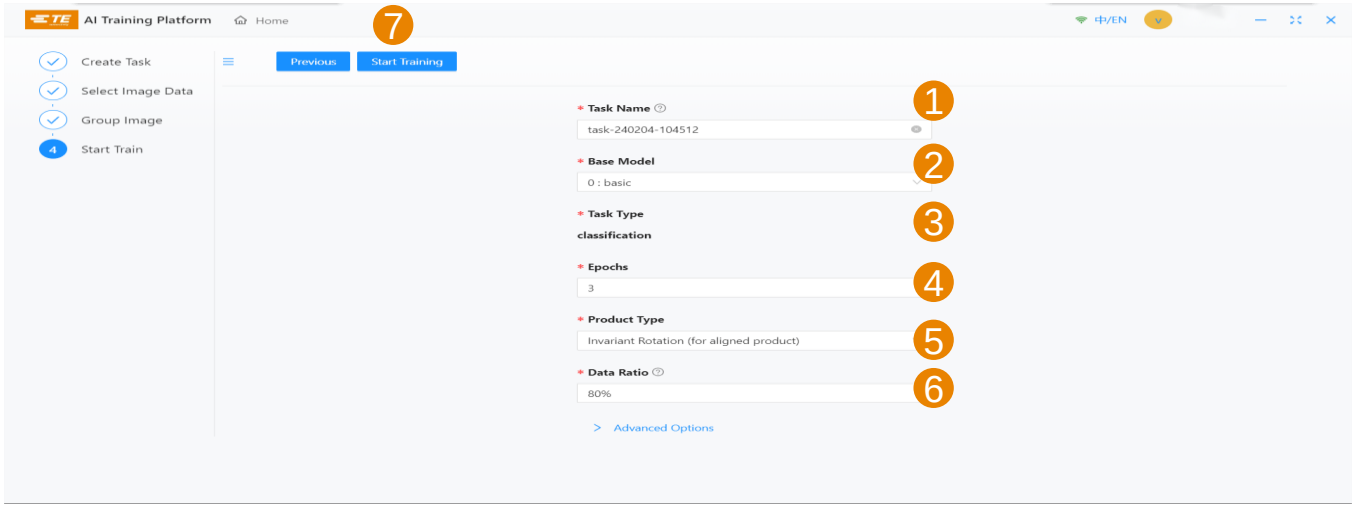
Train the model through the client

Start Training



1. The task name can be edited
2. Confirm the base model and modify it
3. Confirm the task type (non-modifiable)
4. Confirm the training rounds and modify them
5. Select alignment
6. Set the training data set ratio
7. Confirm all parameters and click Start training
8. The training task will be displayed in the task list on the home page, see pages 23-24

Tips:

1. **Alignment is required only for Classification types**
2. **The task type Oneclass is consistent with the Classification step**

Only training data sets are made

Click on the Training Task list page **+Create Dataset** to make the training data set, and follow the creation steps on pages 24-27

Tip: The completed training dataset is displayed in the task list on the home page

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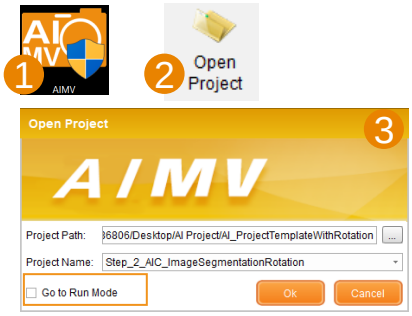
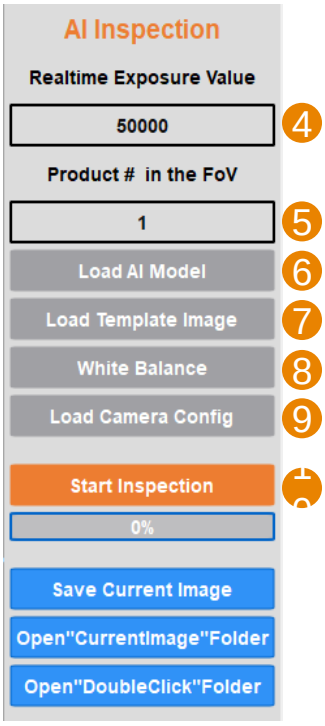
Manual detection deployment

Open Deployment Project



1. Open AIMV Software
2. Click “Open Project”
3. Open Project “Step_5_AIX_XXXXXX”, and check the “Go to Run Mode”
   - Step_5_AIC_Deployment_1Model_NoRotation for “Classification No Rotation”
   - Step_5_AIC_Deployment_1Model_AnyRotation for “Classification Any Rotation”
   - Step_5_AID_Deployment_1Model_AnyRotation for “Defect Location”
   - Step_5_AIA_Deployment_1Model_AnyRotation for “Anomaly Detection”

Start Inspection



4. Input “Realtime exposure Value”, normally it’s the same value in image collection of page 7
5. Input “Product # in the FoV”, this setting is only feasible for “Classification No Rotation”
6. Click “Load AI Model” to load the trained model, then restart the software
7. Click “Load Template Image” to load the template image, normally it’s the same image in page 8 & 9
8. Click “White Balance” if necessary, please refer to Manual Image Collection on the operation
9. Click “Load Camera Config” to load the camera config during image collection
10. After the setting is done, click “Start Inspection” to perform AI inspection on one real product

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On-line detection deployment

TCP Introduction and Hardware Preparation

Please refer to Page 10

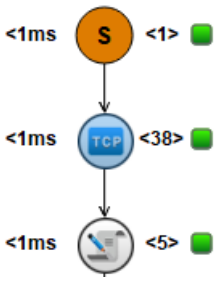
Software Preparation

1. Please go through the **Deployment with Manual Inspection** to make sure the inspection function works correctly

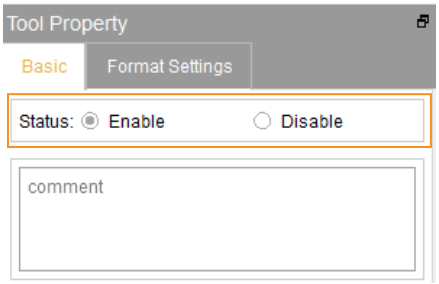
Open Deployment Project

1. Open AIMV Software
2. Click “Open Project”
3. Select Project “Step_5_AIX_XXXXXX”
   - Step_5_AIC_Deployment_1Model_NoRotation for “Classification No Rotation”
   - Step_5_AIC_Deployment_1Model_AnyRotation for “Classification Any Rotation”
   - Step_5_AID_Deployment_1Model_AnyRotation for “Defect Location”
   - Step_5_AIA_Deployment_1Model_AnyRotation for “Anomaly Detection”
4. If the following step is preformed, make sure “Go to Run Mode” is checked to start inline image collection, or else, make sure Go to Run Mode” is unchecked to open the editor mode

Enable PLC/Robot Trigger by TCP

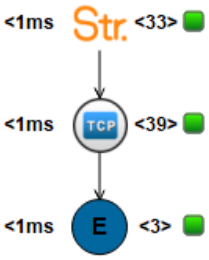


1. Click TCP after S

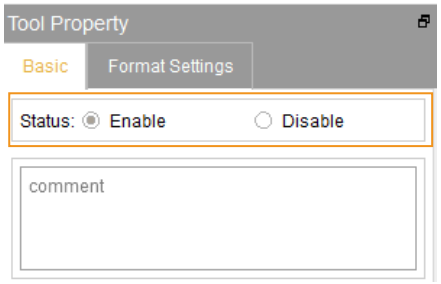


2. Change the tool Status to **Enable** to enable the TCP communication

Enable Feedback to PLC/Robot by TCP



1. Click TCP before E



2. Change the tool Status to **Enable** to enable the TCP communication

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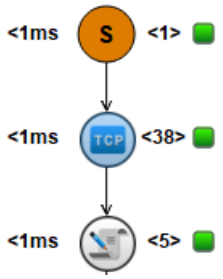
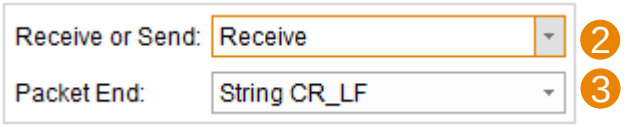
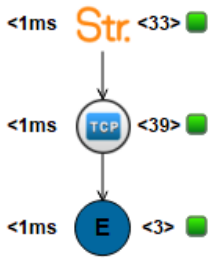
Align the TCP Settings with PLC/Robot



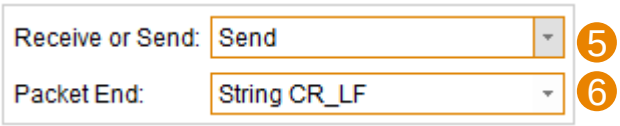
1. Click after



2. Set **Receive or Send** to **Receive** to receive trigger from PLC/robot
3. Make sure the **Packet End** is the same as that of PLC/robot

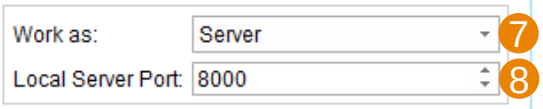


4. Click before



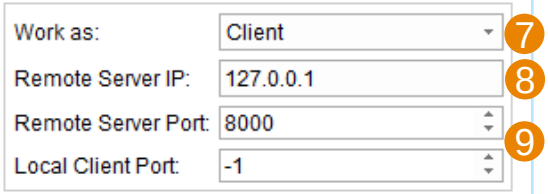
5. Set **Receive or Send** to **Send** to feedback inspection result to PLC/robot
6. Make sure the **Packet End** is the same as that of PLC/robot

AIMV Software works as TCP Server



7. Set the **Work as** to **Server**
8. Set the **Local Server Port** to number between 1001 to 65536, for example 8000
9. Align the TCP settings of PLC/robot with the settings above

AIMV Software works as TCP Client



7. Set the **Work as** to **Client**
8. Set the **Remote Server IP** to PLC/robot IP address
9. Set **Remote Server Port** and **Local Client Port** per the settings of PLC/robot
10. Align the TCP settings of PLC/robot with the settings above

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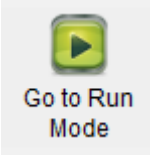
Deployment with Inline Inspection

Start Inline Image Collection

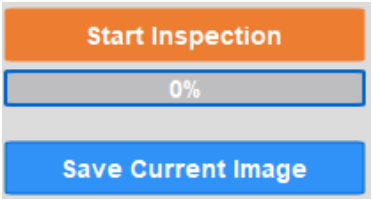


1. Click “Go to Run Mode” and save the settings to start the Run Mode

Inline Inspection with PLC/Robot

PLC/Robot Sends 10 by TCP →

PLC/Robot Sends 1 by TCP →

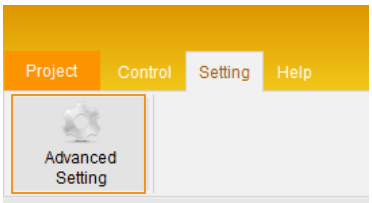


1. If the PLC/Robot Sends 10 by TCP, it’s the same function as click “Start Inspection” button
2. If the PLC/Robot Sends 1 by TCP, it’s the same function as click “Save Current Image” button
3. PLC/Robot will receive 1 if the inspection result is OK
4. PLC/Robot will receive 2 If the inspection result is NG

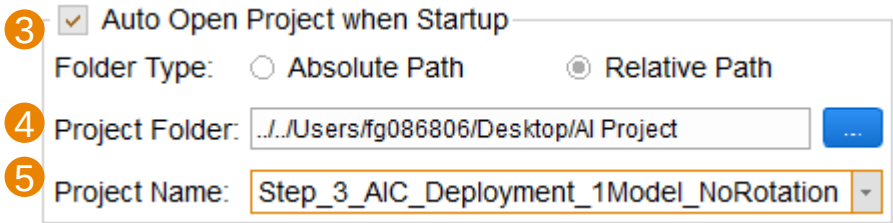
* Please refer to FAQ for connection problem

Auto-Run Inline Deployment when Opening Software

These content is for the customer want to start Image Collection automatically after opening AIMV software



1. Open AIMV Software
2. Click Advanced Setting in Setting Tab



3. Check “Auto Open Project when Startup”
4. Select the Project Folder
5. Select the Project Name to “Step_3_AIX_XXXXXX”, click OK

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Maintenance, Warranty & Copyright

AI Machine Vision is one precise optical instrument which has kinds of advanced technologies in the fields of optics, machine vision, machine learning, etc. Therefore, regular and proper maintenance is very important which will ensure the product works in a good condition and prolong its lifetime.

Maintenance

Before maintenance, the main switch must be turned off first, and make sure the lamphouse is cooled down completely

1. Lens

- The maintenance of lens should be as careful as possible
- Do not use excessive force and scratch action
- Do not move or rotate the lens after one application is deployed
- The lens is usually cleaned when the surface is dirty and influence the usage
- If there is dust and dirt on the surface of lens, clean it by a soft brush, cotton wool, lens paper or wet cotton swabs. Lens paper or cotton swabs can be dipped with very little water so that no water will penetrate into the lens

2. Lighting Unit

- The maintenance of lighting unit should be as careful as possible
- Do not use excessive force and scratch action
- Do not move or rotate the lighting unit after one application is deployed
- If there is dust and dirt around the inner side of the light source, it can be cleaned with soft cloth
- If the LED unit of the light source is damaged or the lighting intensity decreases, it should be replaced in time

3. Camera

- Do not expose the sensor of camera
- Make sure the connection between camera and cable is reliable and stable

4. Camera Cable

- Replace the damaged part of the cable in time to avoid short circuit or cable break

5. Assemble Parts

- When the microscope is not in use, cover it with dust-proof cover

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Warranty

1. Warranty is 1 year from invoice date
2. The seller provides repair and replacement for damaged parts
3. The seller is not be liable for any damage caused by human factors or improper use
4. The seller is not be liable for any damage caused by transportation

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